

# Our Mission

By making industry-leading tools and education available to individuals from all backgrounds, **we level the playing field for future PM leaders.** This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most – Access. We introduce you to industry leaders, **surround you with the right PM ecosystem**, and discover the new world of AI product management skills.

## Global Weather Classification: A Machine Learning Approach

### I. Project Overview

**Objective:** The primary goal of this project is to classify global weather conditions into three categories—Good, Moderate, and Bad—using a range of geographic, atmospheric, and air quality features.

**Motivation:** Accurate weather classification is crucial for agriculture, disaster preparedness, logistics, and climate science. This project explores how machine learning models can automate this classification based on readily available meteorological data.

**Key Approach:** The analysis goes beyond simple model fitting. It incorporates a unique ensemble method (Lyapunov Temperature Probit Ensemble) and includes rigorous statistical testing (McNemar, Wilcoxon) and stress tests to evaluate model robustness and calibration.

### II. Data and Preprocessing

The analysis uses a dataset of global weather observations containing features like temperature, pressure, humidity, wind speed, and various air quality metrics.

#### Problem Reframing & Feature Engineering

**Target Variable:** The original `condition_text` (e.g., "Sunny", "Light rain") was transformed into an ordinal categorical target: 0 (Good), 1 (Moderate), 2 (Bad).

**Feature Selection:** Temperature-related variables (`temperature_fahrenheit`, `feels_like_celsius`) were deliberately excluded to mitigate their role as strong confounders, allowing the model to focus on geographic and atmospheric predictors for better interpretability.

**New Features:** Several engineered features were created to capture complex relationships:

**Statistical Summaries:** Row-wise means, standard deviations, mins, and maxes of all numeric features.

Transformations: Log, inverse, and square transformations of original features.

Dimensionality Reduction: PCA and UMAP components were extracted to capture latent structures.

Clustering: HDBSCAN and Gaussian Mixture Models (GMM) were used to create cluster labels and membership probabilities.

State Modeling: A Gaussian Hidden Markov Model (HMM) was fit to the PCA-transformed data to infer hidden states.

### III. Data Cleaning & EDA

Imputation: Missing values were imputed using the SimpleImputer with the 'most\_frequent' strategy.

Data Split: An 80/20 chronological split was used to simulate a realistic time-series scenario.

Exploratory Analysis:

- skimpy summary showed a wide range of numeric features, with air quality variables (air\_quality\_Carbon\_Monoxide, air\_quality\_PM2.5) exhibiting high standard deviations, indicating skewed distributions.
- A correlation heatmap revealed strong positive correlations between air\_quality\_PM2.5 and air\_quality\_Carbon\_Monoxide.
- A distplot and bar chart showed a highly imbalanced target variable, with most instances labeled as "Good" (0).
- Boxplots confirmed the presence of significant outliers in many air quality features. (See Images for more details)

Modeling and Evaluation: Due to the identified class imbalance, the training data was resampled using ADASYN (Adaptive Synthetic Sampling).

### IV. Individual Model Performance

Five models were trained and evaluated on a held-out test set. LightGBM achieved the highest accuracy and F1-score, while CatBoost achieved the best probability calibration (lowest Brier Score and Log Loss).

| Model | Accuracy | Precision | Recall | F1 Score | Log Loss | Brier Score |
|-------|----------|-----------|--------|----------|----------|-------------|
|-------|----------|-----------|--------|----------|----------|-------------|

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|                     |        |        |        |        |        |        |
|---------------------|--------|--------|--------|--------|--------|--------|
| LightGBM            | 0.9038 | 0.8942 | 0.9038 | 0.8947 | 0.4265 | 0.1689 |
| XGBoost             | 0.8918 | 0.8811 | 0.8918 | 0.8828 | 0.3643 | 0.1683 |
| Random Forest       | 0.8918 | 0.8844 | 0.8918 | 0.8860 | 0.3004 | 0.1643 |
| CatBoost            | 0.8878 | 0.8766 | 0.8878 | 0.8796 | 0.2972 | 0.1637 |
| Logistic Regression | 0.8437 | 0.8549 | 0.8437 | 0.8485 | 0.5214 | 0.2457 |

### 3.2. Lyapunov Temperature Probit Ensemble (LTPE)

A custom ensemble was developed to improve probabilistic calibration. It combines predictions from all five base models using:

- Temperature Scaling: To smooth overconfident probability estimates.
- Probit Transformation: To map probabilities into a latent Gaussian space.
- Lyapunov-Inspired Stability Weighting: Models with lower prediction volatility receive higher weights.
- Weighted Probability Fusion.

Ensemble Performance:

Accuracy: 0.898

F1 Score: 0.889

Log Loss: 0.292 (outperformed all individual models)

Model Weights: The ensemble assigned the highest weight to CatBoost (0.345), followed by Logistic Regression (0.198).

### 3.3. Statistical Testing & Robustness

McNemar's Test: No statistically significant difference ( $p > 0.05$ ) was found between the ensemble's classification decisions and those of the best-performing individual models (LightGBM, CatBoost). This indicates they are statistically equivalent in terms of discrete predictions.

Wilcoxon Signed-Rank Test: Significant differences ( $p < 0.001$ ) were found in the probability distributions of the ensemble versus the individual models, confirming that the ensemble improves probabilistic calibration.

Cliff's Delta (Effect Size): Confirmed a "large" effect size when comparing the ensemble to models with lower accuracy, but no difference or a "negligible" effect when compared to top performers, aligning with the McNemar results.

Quantile Analysis: The ensemble showed better calibration, with error rates dropping to zero in higher-confidence bins, suggesting more reliable probability estimates.

### 3.4. Stress Testing

To assess robustness, models were tested under four data perturbations:

Noise ( $\sigma=0.1$ ): Random Gaussian noise added.

Drift (Factor 1.3): Values scaled.

Missingness (15%): Random features set to zero.

Adversarial ( $\epsilon=0.05$ ): Small perturbations based on random sign.

Robustness Ranking (1 = perfect robustness):

Model Robustness Score

|               |       |
|---------------|-------|
| XGBoost       | 0.869 |
| CatBoost      | 0.863 |
| LightGBM      | 0.838 |
| Ensemble      | 0.820 |
| Random Forest | 0.763 |
| LogReg        | 0.262 |

Conclusion: Gradient boosting models (XGBoost, CatBoost, LightGBM) are significantly more robust to data perturbations than linear models. While the ensemble improves calibration, it does not necessarily provide a robustness advantage over the best boosting models under stressed conditions.

## IV. Impossibility Testing & Explainability

### Impossibility Tests

These tests ensure the model has truly learned a meaningful pattern:

Label Randomization: When the target labels were shuffled, all models' performance dropped to near chance levels (~0.33 accuracy for a 3-class problem), confirming that the models are not memorizing spurious patterns.

Feature Projection & PCA Bottleneck: Performance decreased but remained above chance when features were projected onto random bases or reduced to a single PCA component, indicating a moderate degree of redundancy in the feature space.

Noise Dominance: Increasing noise to high levels destroyed most predictive power, as expected.

#### Model Explainability (SHAP & Permutation Importance)

Global Importance: Permutation importance for the best-performing models (CatBoost, XGBoost) identified latitude, longitude, gmm\_ and hdbscan\_ cluster features, and several air quality variables as the most influential predictors.

Local Explanations (SHAP): SHAP summary plots for tree-based models showed that being in a certain cluster or region (e.g., cluster 0 or 2) can push predictions towards 'Good', while higher values of air pollutants like air\_quality\_Sulphur\_dioxide and air\_quality\_PM2.5 push predictions towards 'Bad'.

#### V. Conclusion

Model Selection is a Trade-off: LightGBM is the best choice for maximizing discriminative performance (accuracy/F1), while CatBoost is superior for probabilistic calibration (log loss/Brier score). The choice depends on the application's needs.

Ensemble for Calibration: The proposed Lyapunov Temperature Probit Ensemble significantly improves probability calibration without a statistically significant gain in classification accuracy. Its primary value lies in creating more reliable uncertainty estimates.

Robustness is Key: Gradient boosting models are far more resilient to data distribution shifts than simpler models. XGBoost demonstrated the highest overall robustness.

Model Reliability: Impossibility tests confirm that the models are learning genuine relationships, not memorizing noise.



